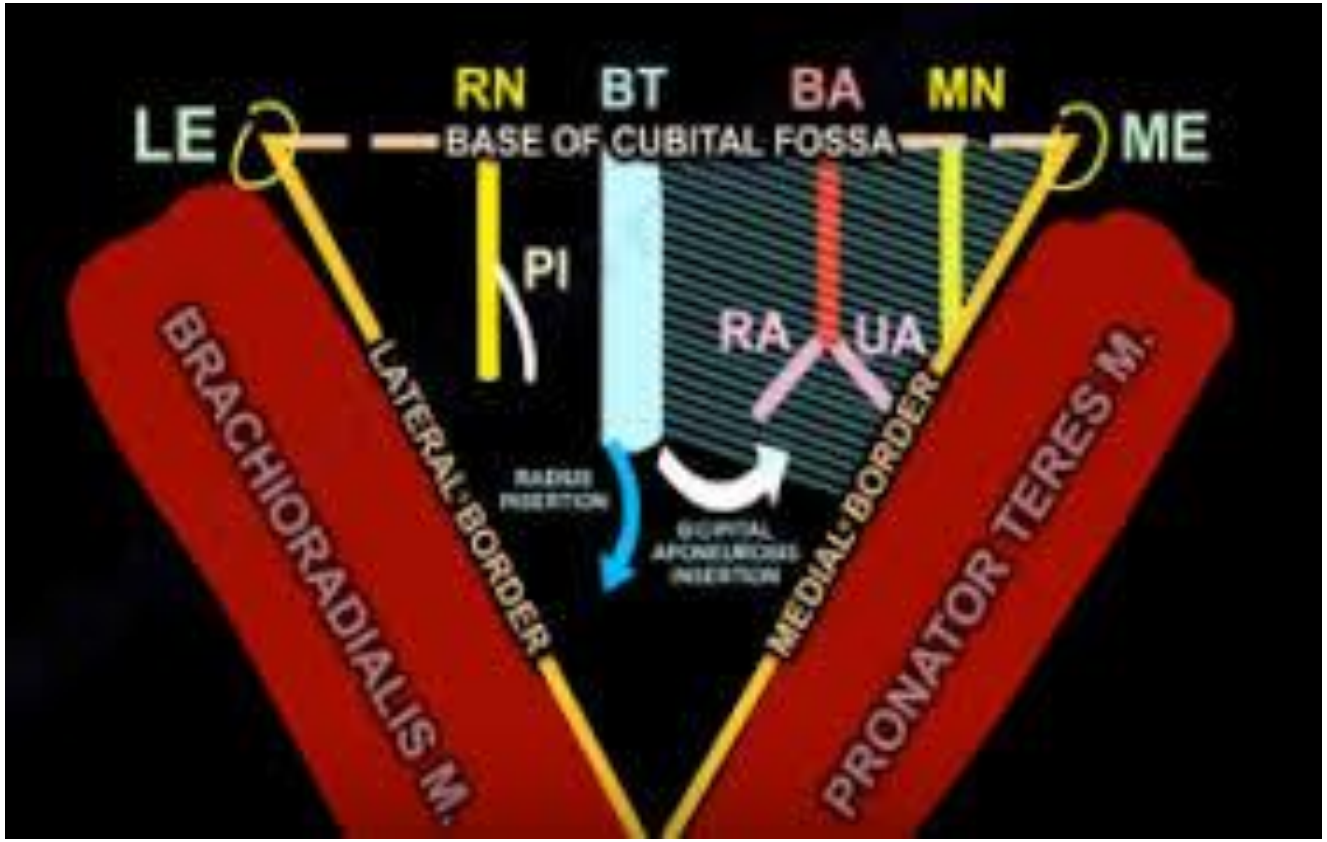


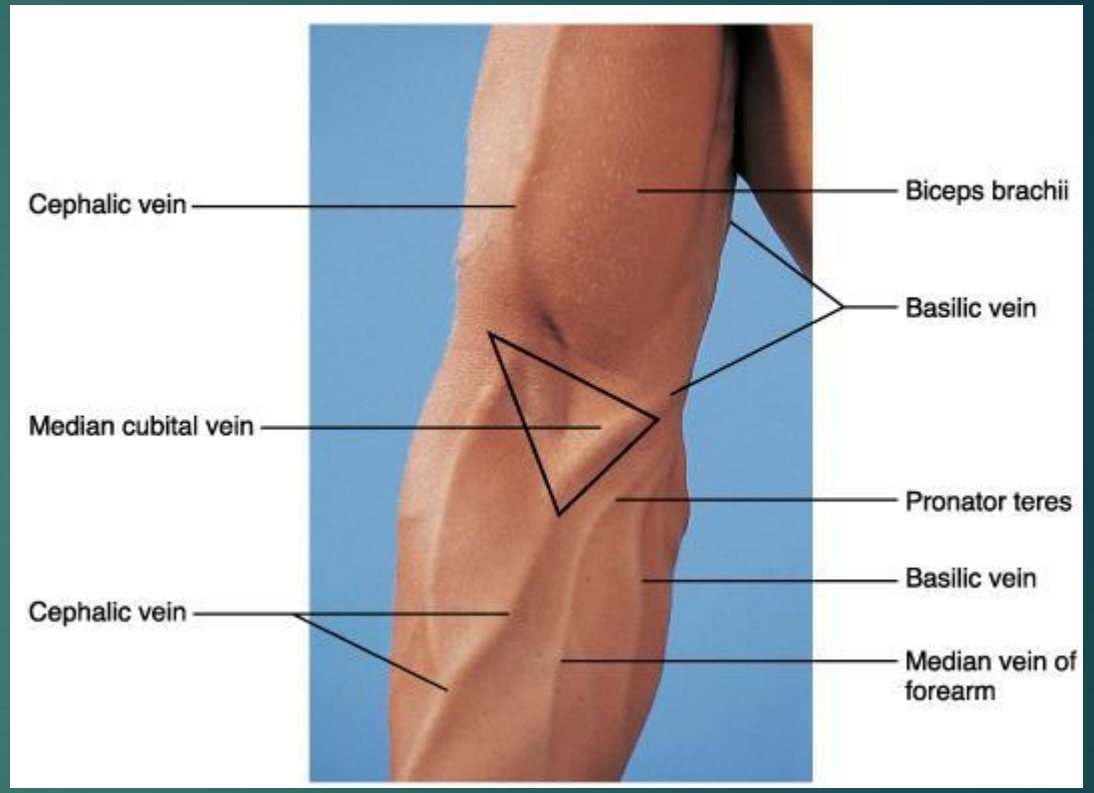
Cubital fossa

200 LEVEL MBBS/BDS

LASUCOM

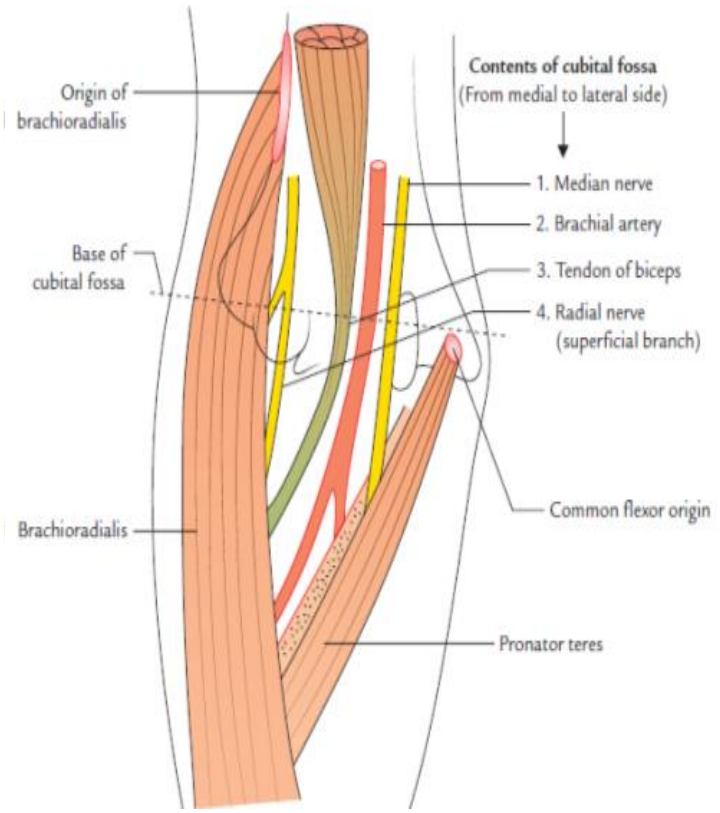
Introduction

- The cubital fossa is a skin depression that lies in front of the elbow and is triangular in shape.
- It corresponds (i.e., homologous) to the popliteal fossa of the lower limb.



BOUNDARIES

- *Lateral:* Medial border of brachioradialis muscle.
- *Medial:* Lateral border of pronator teres muscle.
- *Base:* An imaginary horizontal line, joining the front of two epicondyles of the humerus.
- *Apex:* Meeting point of the lateral and medial boundaries.



BOUNDARIES(Cont'd)

- *Floor:* It is formed by two muscles, *brachialis* in the upper part and *supinator* in the lower part.
- *Roof: Skin and superficial(containing (i) median cubital vein connecting cephalic and basilic veins, and medial and lateral cutaneous nerves of forearm) and deep fasciae which is strengthened by aponeurosis of biceps muscle)*

.

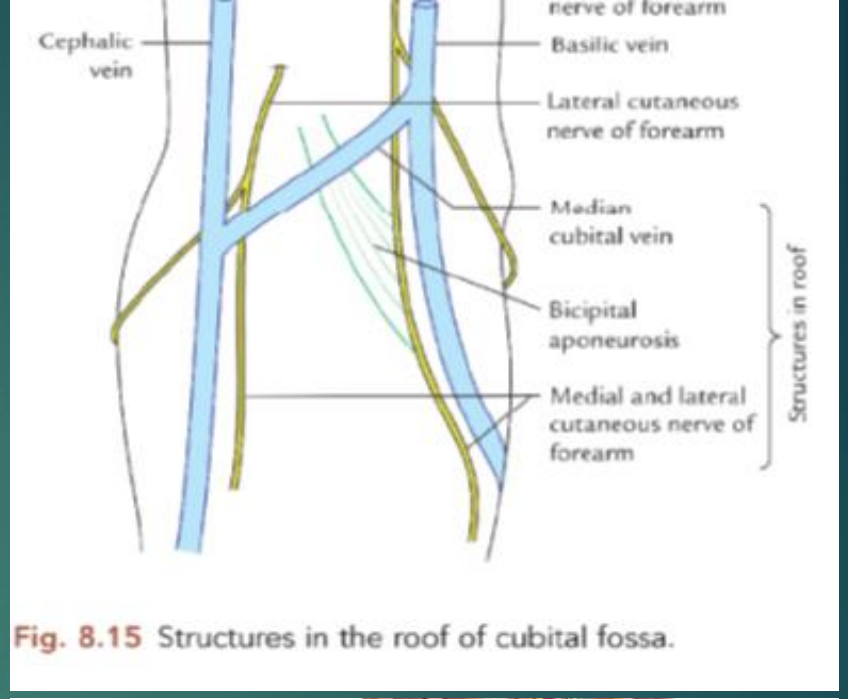
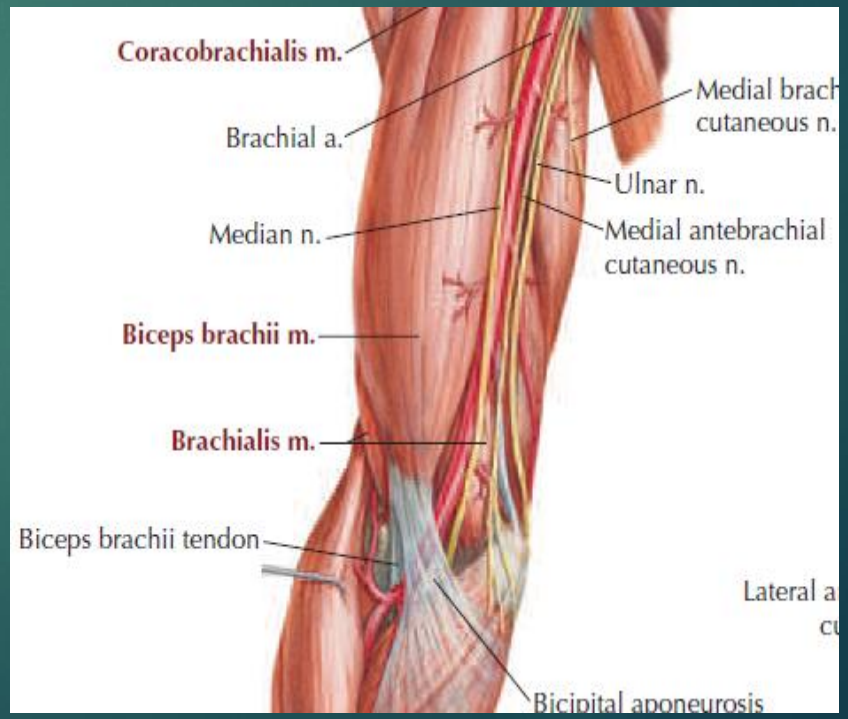
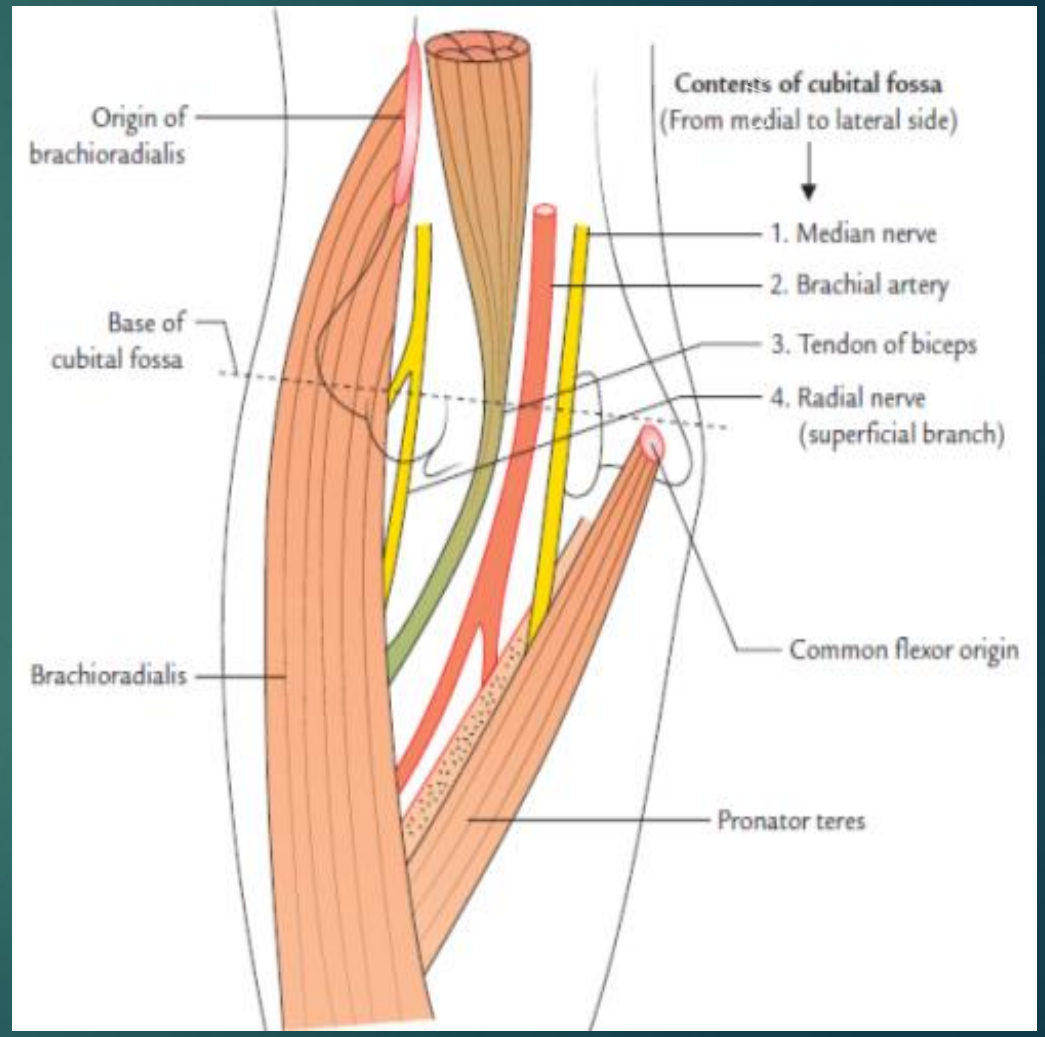


Fig. 8.15 Structures in the roof of cubital fossa.



CONTENTS

- From medial to lateral, the cubital fossa contains
- the **M**edian nerve,
- the bifurcation of the **B**rachial artery into the ulnar and the radial arteries,
- the tendon of the **B**iceps muscle,
- and the radial nerve and its superficial branch.



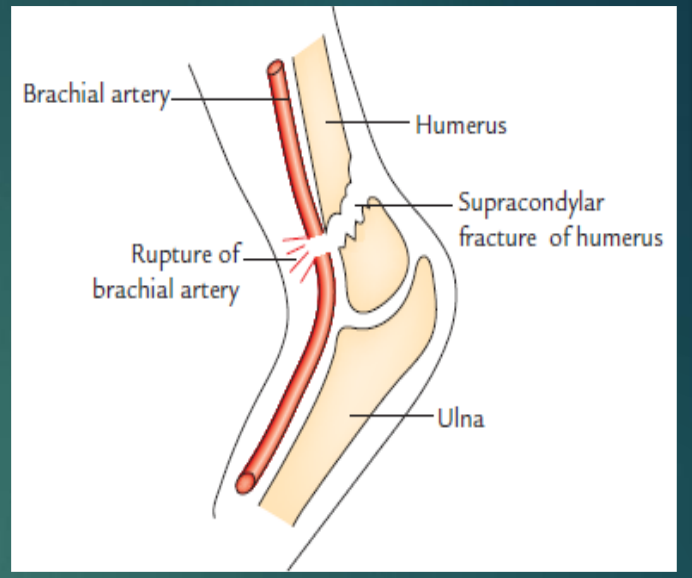
Clinical correlation

Clinical significance of cubital fossa: The knowledge of anatomy of the cubital fossa is clinically important for the following reasons:

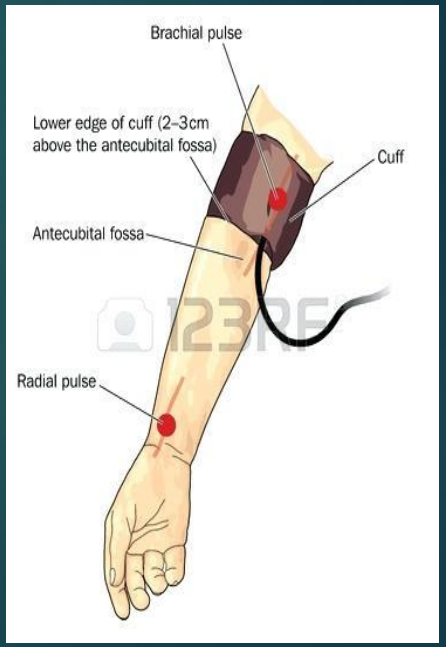
(a) The **median cubital vein** in this region is the vein of choice for collecting blood samples and giving intravenous injections.

(b) The **brachial pulse** in this region is easily felt medial to biceps tendon, for **recording the blood pressure.**

(c) To deal with the fractures around elbow, viz. supracondylar fracture of the humerus. The contents of cubital fossa especially the brachial artery and median nerve are vulnerable in supracondylar fracture of the humerus.



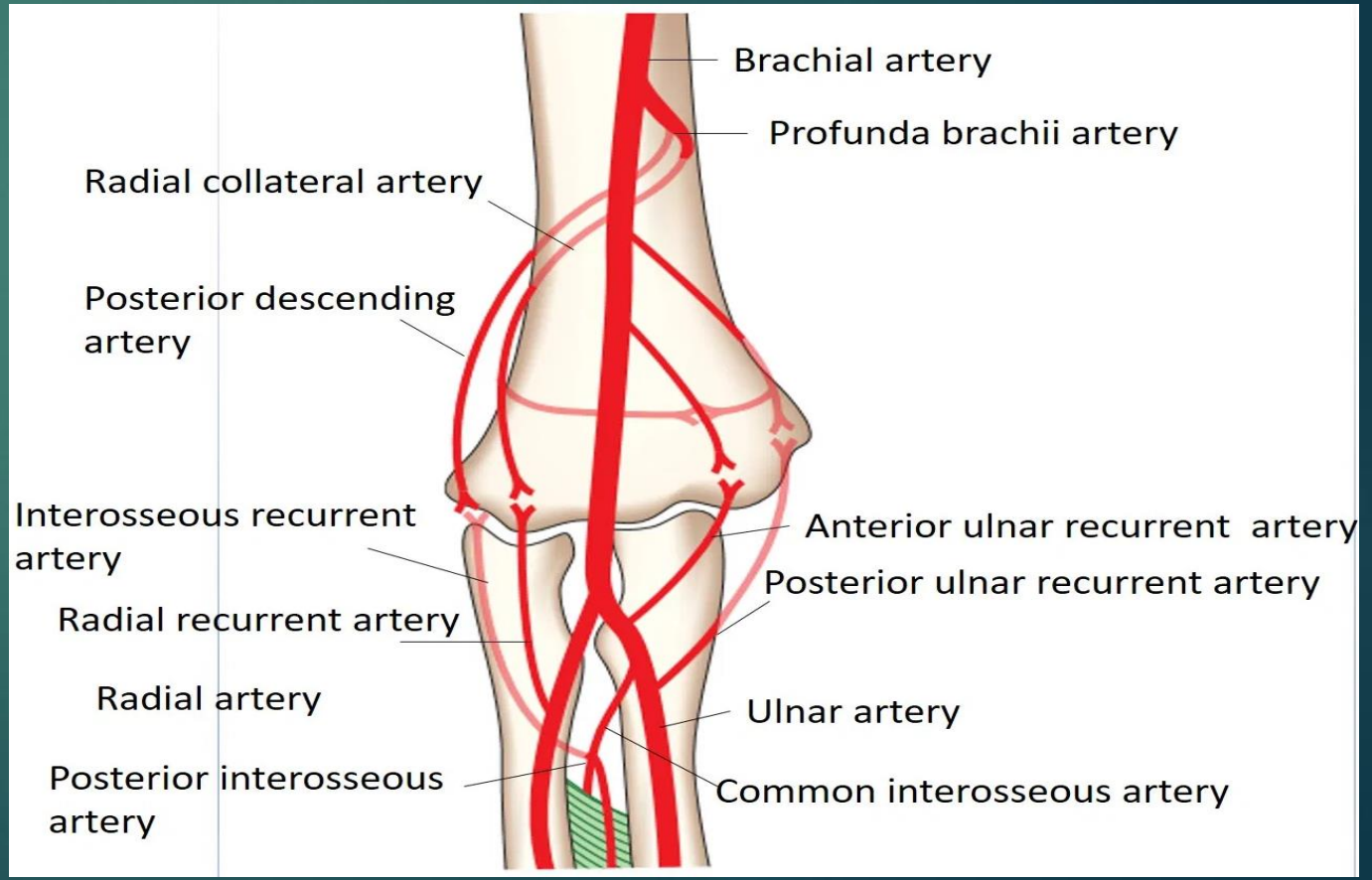




ELBOW ANASTOMOSIS

- The periarticular **arterial anastomosis of the elbow** consists of several arteries that supply the elbow joint and its supporting structures.
- It functions to allow blood to flow around the elbow joint no matter which position the joint is in.
- The arteries that contribute include:

1. From the brachial artery
   - superior ulnar collateral artery
   - inferior ulnar collateral artery
2. From the deep brachial artery
   - middle collateral artery
   - radial collateral artery
3. From the radial artery
   - radial recurrent artery
4. From the ulnar artery
   - common interosseous artery
     - anterior interosseous artery
     - posterior interosseous artery
       - interosseous recurrent artery
   - posterior ulnar recurrent artery
   - anterior ulnar recurrent artery



ELBOW ANASTOMOSIS

1. Anterior to medial epicondyle:

Inferior ulnar collateral artery and branch from the superior ulnar collateral artery (branches of the brachial artery), anastomose with anterior ulnar recurrent artery (branch of the ulnar artery)

2. Posterior to medial epicondyle:

Superior ulnar collateral artery and a branch from the inferior ulnar collateral artery (branches of brachial artery), anastomose with posterior ulnar recurrent artery (branch of the ulnar artery)

3. Anterior to lateral epicondyle:

Radial collateral artery (anterior descending branch of the profunda brachii artery), anastomose with radial recurrent artery (branch of the radial artery)

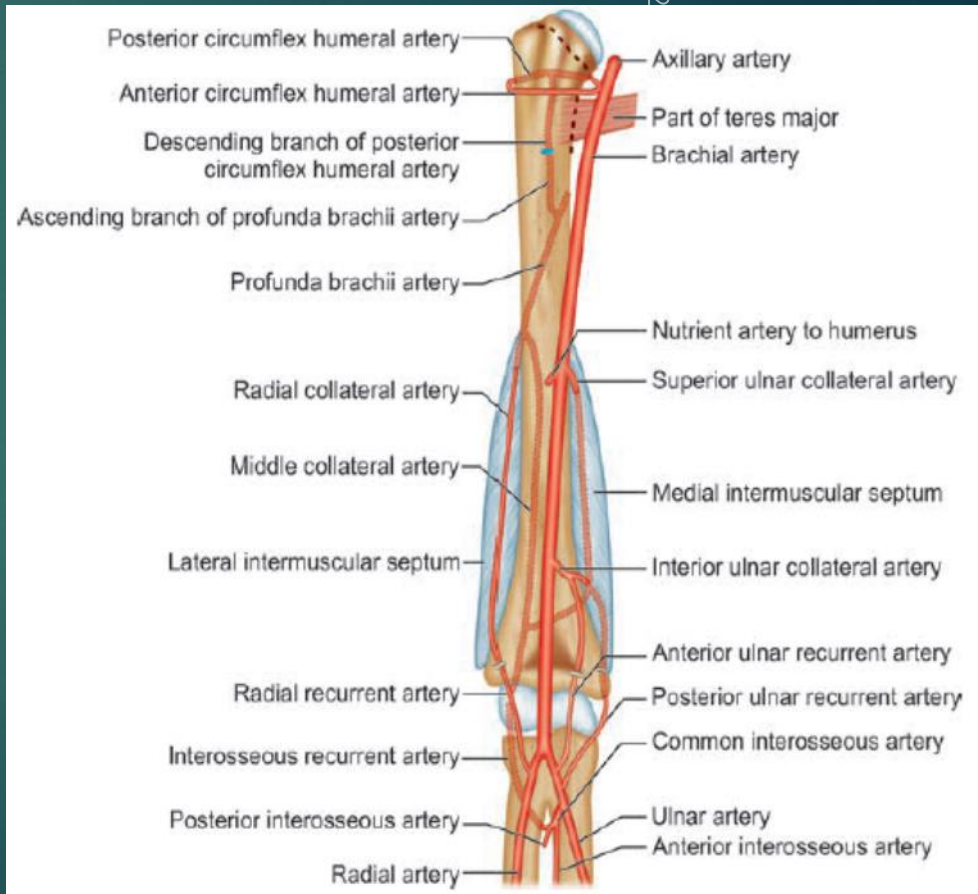


Fig. 56: Elbow anastomosis

ELBOW ANASTOMOSIS

4. Posterior to lateral epicondyle:

Middle collateral artery (posterior descending branch of the profunda brachii artery), anastomose with recurrent interosseous artery (branch of the posterior interosseous artery, a branch of common interosseous artery, a branch of the ulnar artery)

5. Superior to olecranon fossa:

Middle collateral artery (branch of the profunda brachii artery), anastomose with transverse branch from the posterior division of the inferior ulnar collateral artery

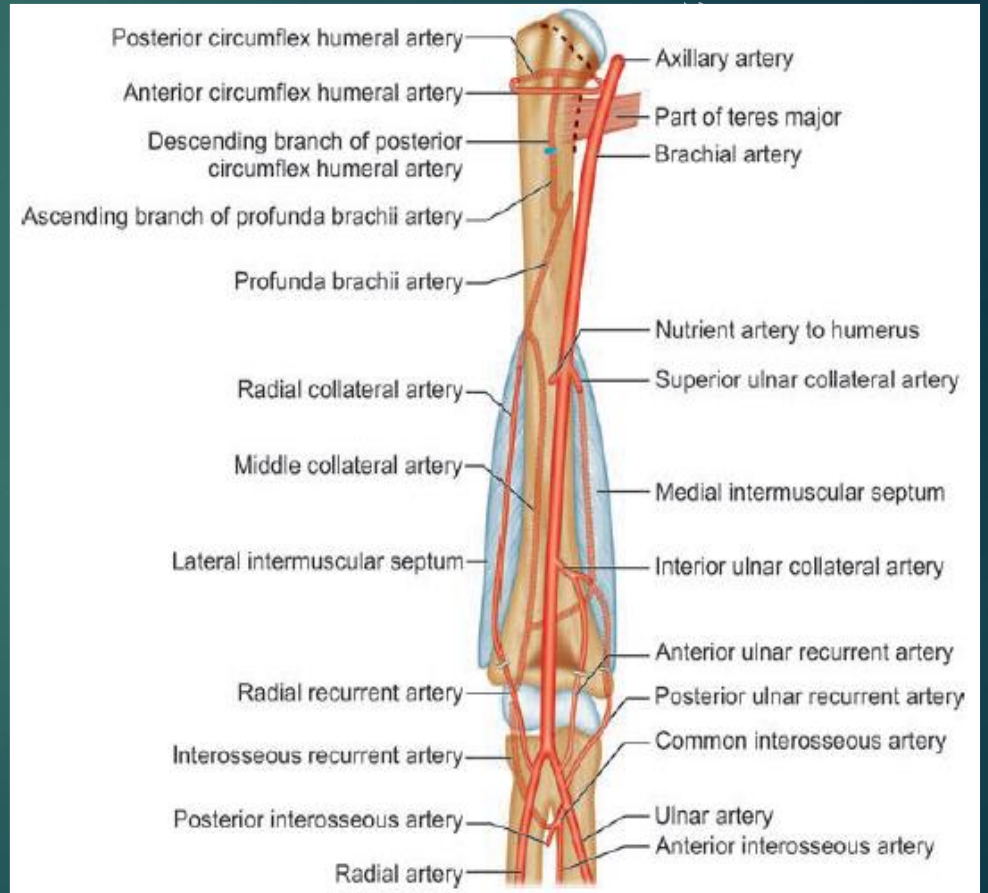
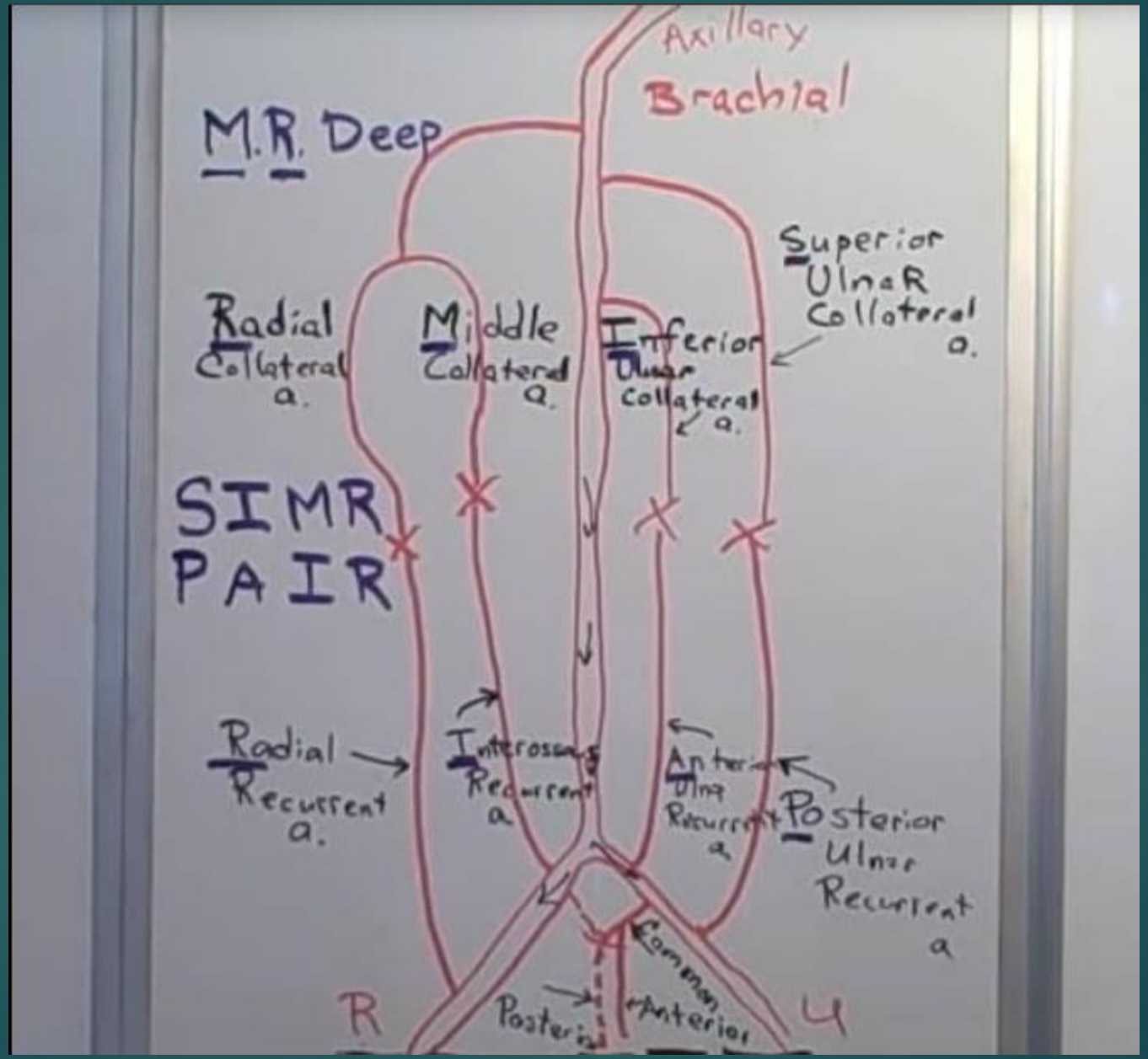


Fig. 56: Elbow anastomosis

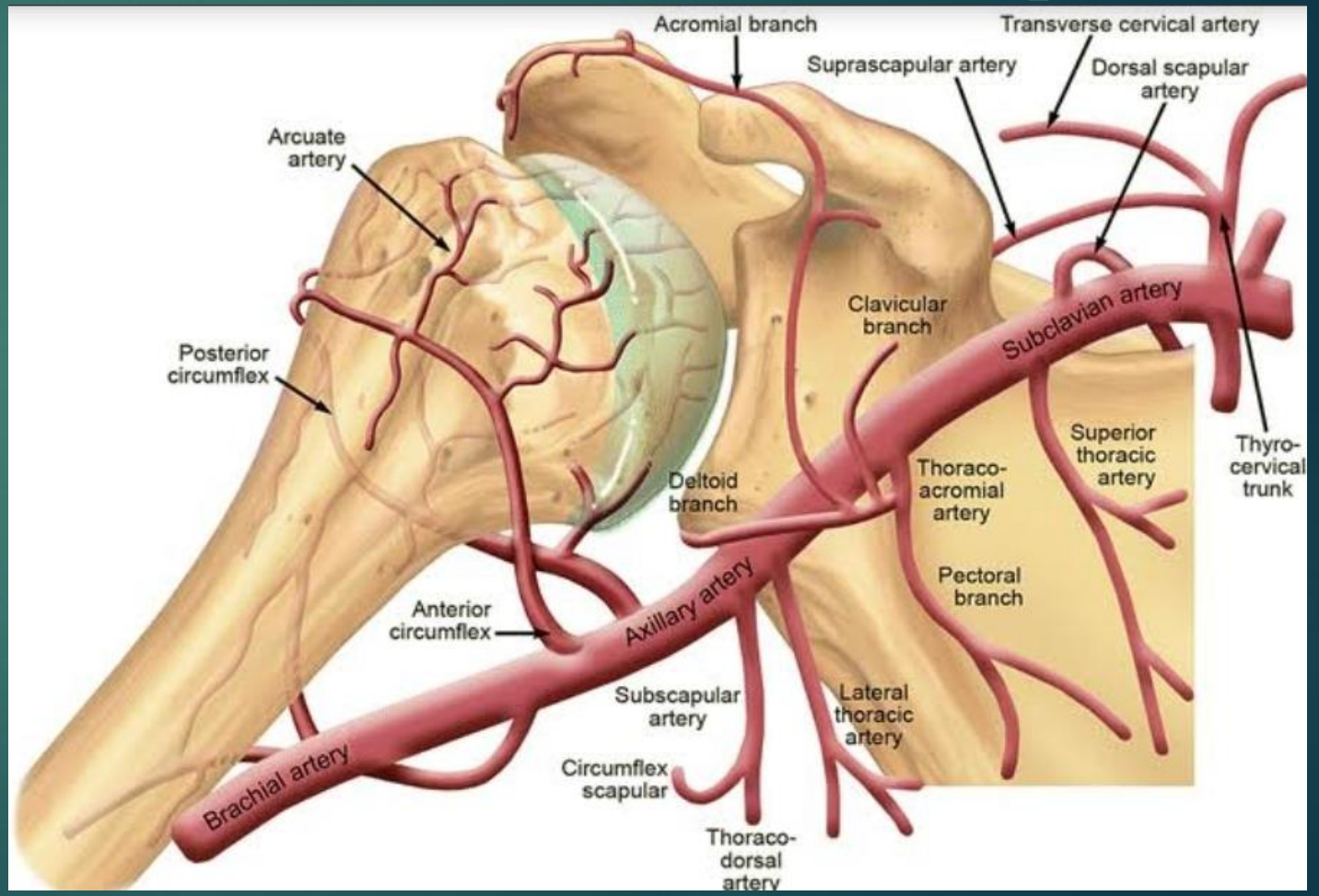

Axillary
Brachial
M.R. Deep
Radial Collateral a.
Middle Collateral a.
Inferior Ulnar Collateral a.
Superior Ulnar Collateral a.
SIMR PAIR
Radial Recurrent a.
Interosseous Recurrent a.
Anterior Ulnar Recurrent a.
Posterior Ulnar Recurrent a.
Common
R
Posterior
Anterior
U

ANASTOMOSIS AROUND THE SURGICAL NECK OF HUMERUS

- Formed by

1. Anterior circumflex humeral artery
2. Posterior circumflex humeral artery and
3. Ascending branch of profunda brachii artery

- The first two are from the axillary artery while the third artery is a branch of the brachial artery.




thank
you